Machine Learning

31-03-26

Linear Regression

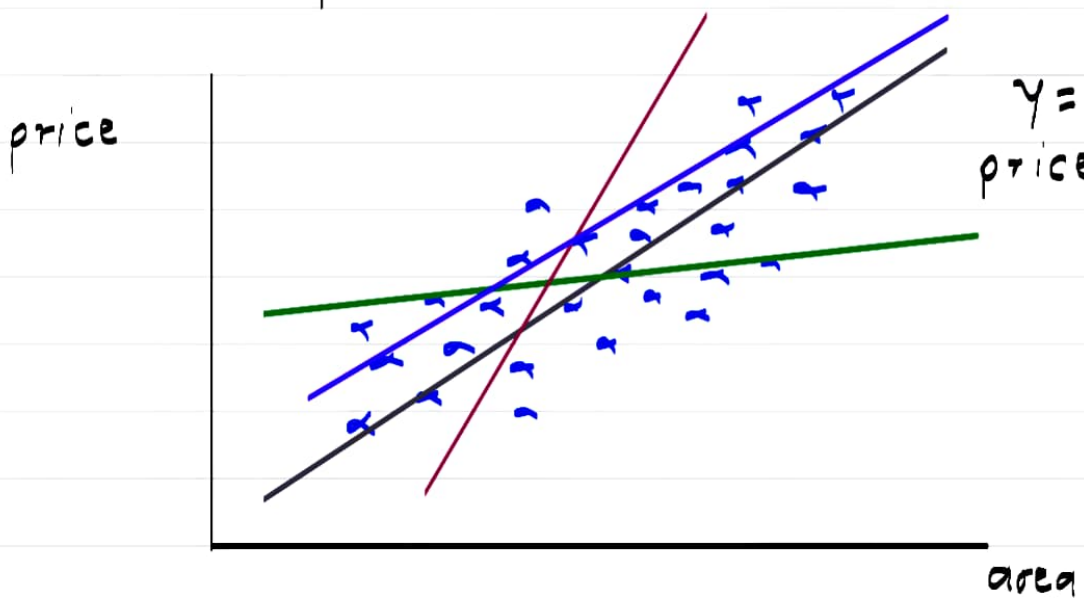
- supervised ML model
- used only for regression
- parametric model

Simple Linear Regression

- has only one feature as input
- Linear means → this model tries to learn linear relationship between feature and target

eg.

house area	house price
1200	21
1400	21.5
1600	24
2000	26
⋮	⋮

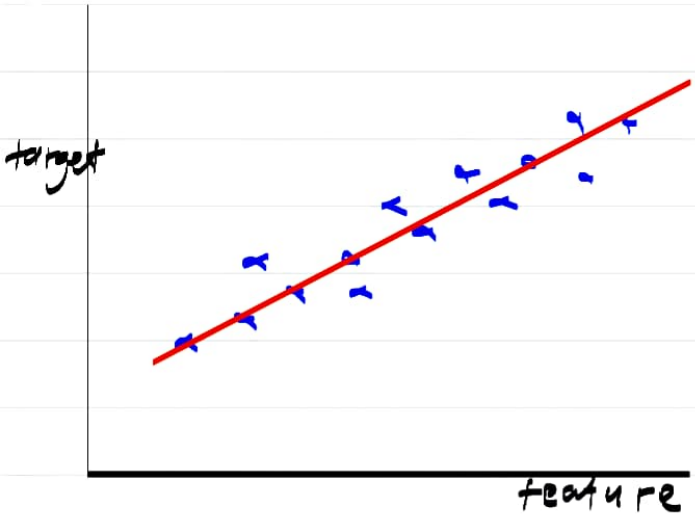


$y = f(x)$

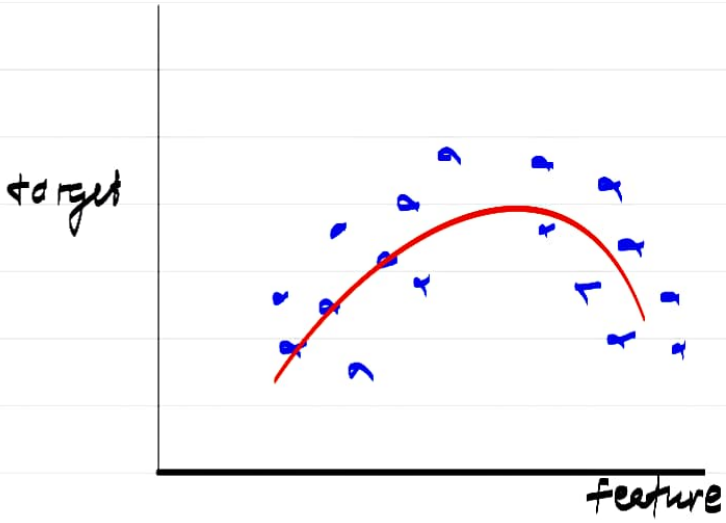
price = f(area)

- when the data has relationship like this which can be approximately explained by a straight line then we use simple linear regression.

- Linear regression assumes that there is an approximate linear relationship between feature and target.



here linear regression would work



here linear regression won't work at all.

- Objective of linear regression model training is to find the straight line which is best to capture linear relationship among features and target.

supervised ML objective

↓

learn relationship betn features and target

↓

linear regression

↓

learn linear relationship.